

[Figure 7.2](#) also uses simulated price data and shows that there can be a scenario where the past 12 month excess performance is positive, but because of a sharp turn in recent performance, the MA rule is negative. Of course, in general, MA and TMOM point in the same direction, however, [Figures 7.1](#) and [7.2](#) highlight that this conclusion is not inevitable.

As the equations highlight, TMOM and MA rules are tied together: Time series momentum rules (TMOM) are a function of moving average (MA) rules. And while TMOM and MA triggers are highly correlated, there are circumstances where the rules have a difference of opinion (see [Figures 7.1](#) and [7.2](#)). Our analysis suggests that this “difference of opinion,” where one rule zigs while the other zags, creates a trading rule (ROBUST) that is more resilient than either trading rule as a stand-alone risk management device.

How Does ROBUST Work?

In this following section, we show performances for the simplest risk-management models that seem to work: MA, MOM, and ROBUST.² We look at how each of these systems performs on the five asset classes outlined below:

- SPX = S&P 500 Total Return Index (US Stocks)
- EAFE = MSCI EAFE Total Return Index (International Stocks)
- REIT = FTSE NAREIT All Equity REITS Total Return Index (Real Estate)
- GSCI = S&P GSCI Total Return CME (Commodities)
- LTR = The Merrill Lynch 10-year US Treasury Futures Total Return Index (Bonds)

The results presented are gross, and no fees are included. All returns are total returns and include the reinvestment of distributions (e.g., dividends). Proceeds are invested in treasury bills when a risk management rule is triggered.

Buy-and-hold results for the five asset classes from January 1976 to December 2014 are presented in [Table 7.3](#).